

SSS 1-3 Smart Irrigation Demonstrator Flashcards 941-960

941. SMART IRRIGATION	Smart irrigation uses measurements, rules, models, or predictions to adapt water...
942. STATE VARIABLE	A state variable describes a system condition needed to represent current status...
943. CONTROL VARIABLE	A control variable is an adjustable operating decision such as valve state, irri...
944. SETPOINT	A setpoint is the target value that a control system attempts to maintain or ach...
945. THRESHOLD CONTROL	Threshold control triggers an action when a measured variable crosses a defined ...
946. HYSTERESIS	Hysteresis uses different on and off thresholds to reduce rapid switching around...
947. FEEDBACK CONTROL	Feedback control measures system response and uses that information to adjust la...
948. FEEDFORWARD CONTROL	Feedforward control anticipates a disturbance or demand using information measur...
949. SOIL-MOISTURE INPUT	Soil-moisture measurements can inform irrigation decisions when sensor placement...
950. WEATHER INPUT	Weather observations can help estimate atmospheric water demand and recent water...
951. CROP-STAGE INPUT	Crop development stage influences rooting, canopy, sensitivity, and water demand...
952. FLOW VERIFICATION	Flow measurement verifies whether commanded irrigation actually delivered water ...
953. TANK-LEVEL CONSTRAINT	Tank level can constrain irrigation decisions when stored water is insufficient ...
954. FAULT DETECTION	Fault detection identifies abnormal behavior such as no flow after valve opening...
955. FAIL-SAFE STATE	A fail-safe state is the predefined safe operating condition used when critical ...
956. DECISION LOGIC	Decision logic combines states, thresholds, priorities, constraints, and actions...
957. PERFORMANCE KPI	A key performance indicator quantifies control performance, such as water applie...
958. RULE-BASED BENCHMARK	A benchmark controller provides a reference strategy against which a more advanc...
959. CONTROL SCENARIO	Control scenarios test system behavior under different water supply, demand, sen...
960. SMART IRRIGATION EVALUATION	Evaluation compares water use, crop-zone conditions, reliability, faults, and op...